

LAPORAN PRAKTIKUM 10 - PENGEMBANGAN WEB SERVICE

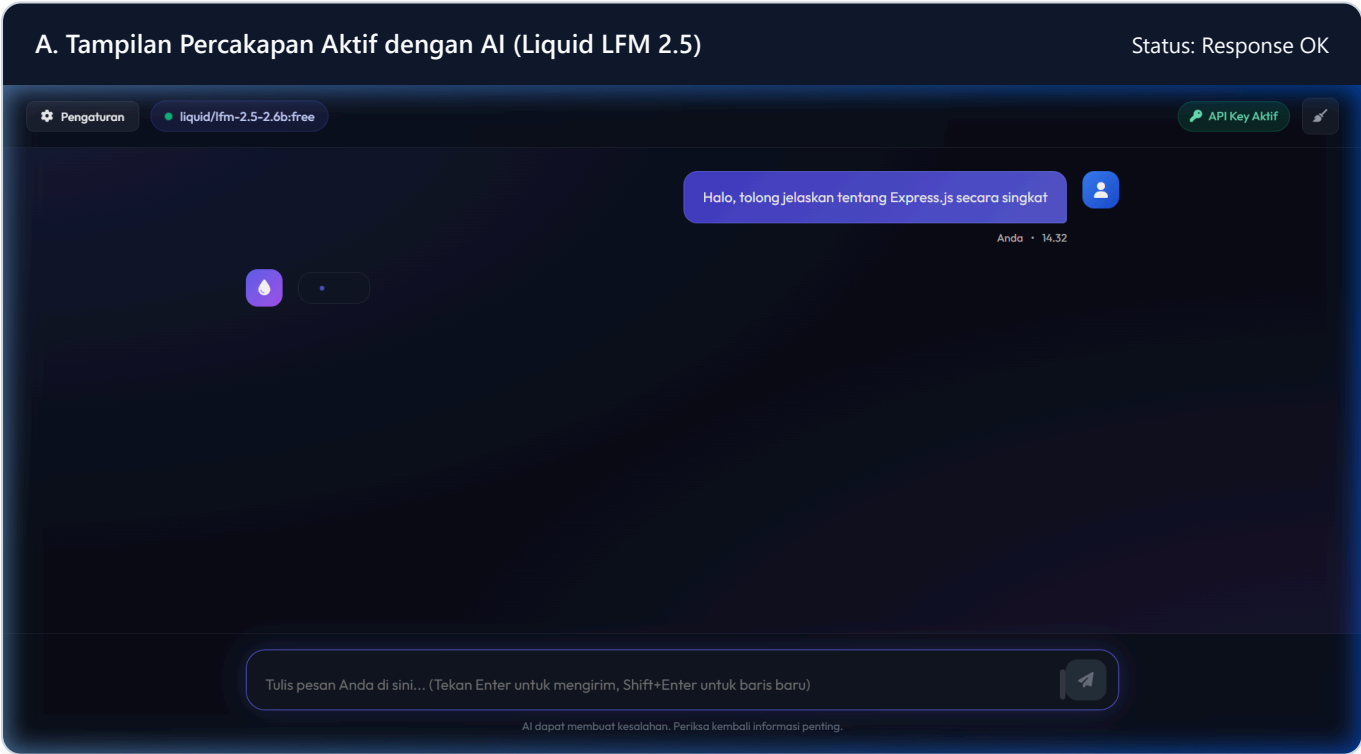
Implementasi Express.js AI Chatbot Menggunakan Model Liquid LFM 2.5 (OpenRouter API)

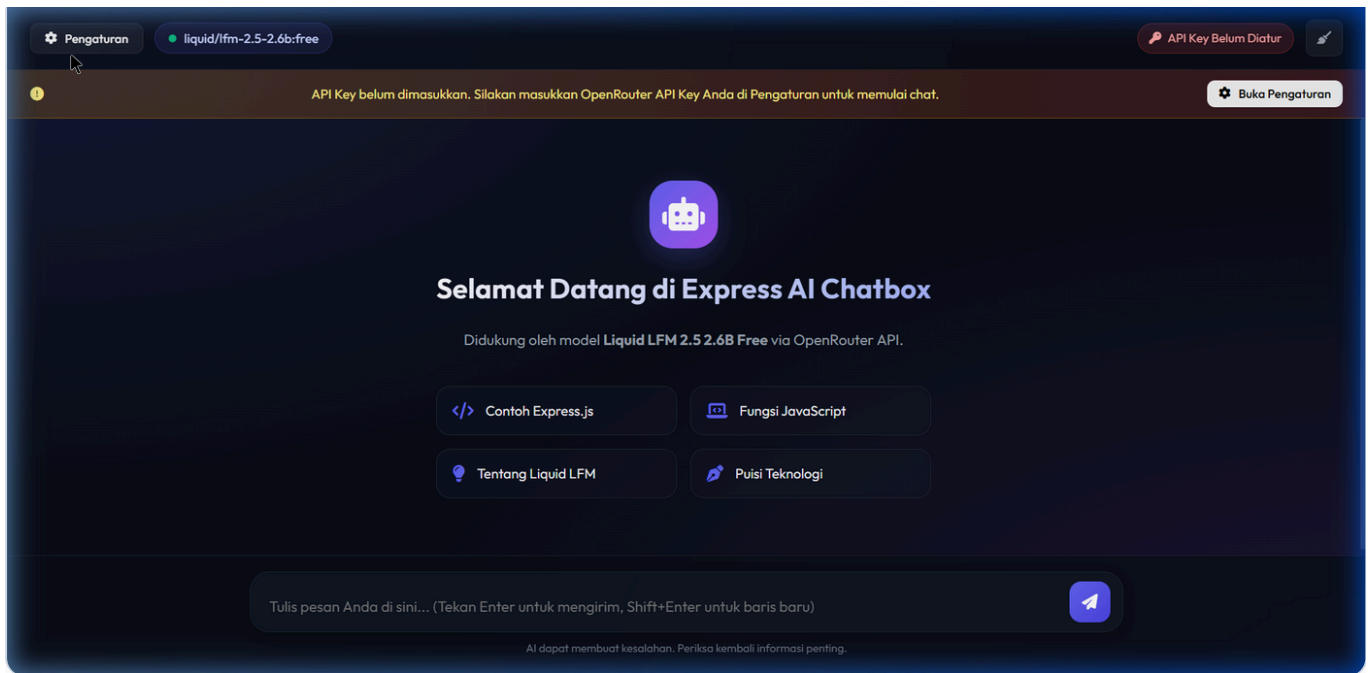
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NIM	20240140118
Praktikum / Matkul	Pengembangan Web Service (Praktikum 10)
Link Repositori GitHub	https://github.com/ilham101106/118_ChatboxAI.git

1. Deskripsi Pengerjaan Tugas

Pada praktikum 10 ini, saya membuat aplikasi web chatbox interaktif berbasis **Node.js & Express.js**. Server Express bertindak sebagai backend penyedia API yang menghubungkan aplikasi web ke model AI **Liquid LFM 2.5 2.6B Free** (liquid/lfm-2.5-2.6b:free) melalui OpenRouter API. Antarmuka dibuat responsif dengan fitur *collapsible drawer settings*, Markdown formatting, serta rendering blok kode interaktif.

2. Hasil Tangkapan Layar (Screenshots UI & Chat AI)





3. Kode Program Utama (Easy CodeSnap Style)

Berikut adalah potongan source code program yang disusun dengan format CodeSnap:

```
public/index.html HTML5

1 <!DOCTYPE html>
2 <html lang="id">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Express AI Chatbox - Liquid LFM 2.5</title>
7
8   <!-- SEO & Favicon -->
9   <meta name="description" content="Aplikasi Chatbot AI berbasis Express.js menggunakan model liquid/lfm-
10 2.5-2.6b:free dari OpenRouter.">
11
12   <!-- Google Fonts: Outfit & Fira Code -->
13   <link rel="preconnect" href="https://fonts.googleapis.com">
14   <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
15   <link href="https://fonts.googleapis.com/css2?family=FiraCode:wght@400;500;600&family=Outfit:wght@300;
16 400;500;600;700&display=swap" rel="stylesheet">
17
18   <!-- FontAwesome Icons -->
19   <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.cs
20 s">
21
22   <!-- Highlight.js CSS for Code Highlighting -->
23   <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/highlight.js/11.9.0/styles/tokyo-ni
24 ght-dark.min.css">
25
26   <!-- Main Stylesheet -->
27   <link rel="stylesheet" href="style.css">
28 </head>
29 <body>
30   <div class="app-container" id="appContainer">
31
32     <!-- Sidebar / Drawer Panel (Dapat Disembunyikan / Toggle Desktop & Mobile) -->
33     <aside class="sidebar" id="sidebar">
34       <div class="sidebar-header">
35         <div class="brand">
36           <div class="brand-icon">
37             <i class="fa-solid fa-droplet"></i>
38           </div>
39           <div class="brand-text">
40             <h2>Liquid Express</h2>
41             <span class="badge">AI Chatbot</span>
42           </div>
43         </div>
44         <button class="btn-icon hide-sidebar-btn" id="closeSidebarBtn" title="Sembunyikan Sidebar">
45           <i class="fa-solid fa-chevron-left"></i>
46         </button>
47       </div>
48       <div class="sidebar-content">
49
50         <!-- API Key Configuration Card (Ops 2) -->
51         <div class="config-card">
52           <div class="config-title">
```

```

51         <i class="fa-solid fa-key"></i>
52         <span>OpenRouter API Key</span>
53         <span class="status-indicator" id="apiKeyStatusDot"></span>
54     </div>
55     <p class="config-desc">API Key disimpan secara aman di <em>Local Storage</em> browser Anda.</p>
56
57     <div class="input-group">
58         <div class="password-wrapper">
59             <input type="password" id="apiKeyInput" placeholder="sk-or-v1-..." autocomplete="off">
60             <button class="btn-toggle-eye" id="toggleApiKeyVisibility" title="Lihat/Sembunyikan Key">
61                 <i class="fa-solid fa-eye"></i>
62             </button>
63         </div>
64         <button class="btn btn-primary" id="saveApiKeyBtn">
65             <i class="fa-solid fa-floppy-disk"></i> Simpan Key
66         </button>
67     </div>
68
69     <a href="https://openrouter.ai/keys" target="_blank" rel="noopener noreferrer" class="link-external">
70         <i class="fa-solid fa-arrow-up-right-from-square"></i> Dapatkan API Key Gratis di OpenRouter
71     </a>
72 </div>
73
74 <!-- Model Selection -->
75 <div class="config-card">
76     <div class="config-title">
77         <i class="fa-solid fa-microchip"></i>
78         <span>Model AI</span>
79     </div>
80     <div class="select-wrapper">
81         <select id="modelSelect">
82             <option value="liquid/lfm-2.5-2.6b:free" selected>Liquid LFM 2.5 2.6B (Free)</option>
83             <option value="meta-llama/llama-3.3-70b-instruct:free">Llama 3.3 70B (Free)</option>
84             <option value="google/gemini-2.0-flash-exp:free">Gemini 2.0 Flash (Free)</option>
85             <option value="deepseek/deepseek-r1:free">DeepSeek R1 (Free)</option>
86             <option value="qwen/qwen-2.5-coder-32b-instruct:free">Qwen 2.5 Coder 32B (Free)</option>
87         </select>
88     </div>
89 </div>
90
91 <!-- System Prompt Setting -->
92 <div class="config-card">
93     <div class="config-title">
94         <i class="fa-solid fa-sliders"></i>
95         <span>System Prompt</span>
96     </div>
97     <textarea id="systemPromptInput" rows="3" placeholder="Atur karakter AI... (Contoh: Kamu adalah asisten pemrograman yang ramah dan membantu.)">Kamu adalah Liquid AI, asisten virtual yang cerdas, ramah, dan sangat membantu dalam bahasa Indonesia.</textarea>
98     <button class="btn btn-secondary btn-sm" id="resetSystemPromptBtn">
99         <i class="fa-solid fa-rotate-left"></i> Reset Prompt
100     </button>
101 </div>
102
103 <!-- Actions -->
104 <div class="sidebar-actions">
105     <button class="btn btn-danger w-100" id="clearChatBtn">
106         <i class="fa-solid fa-trash-can"></i> Hapus Percakapan
107     </button>
108 </div>
109
110 </div>
111
112 <!-- Footer Info -->
113 <div class="sidebar-footer">

```

```

114     <p><i class="fa-solid fa-server"></i> Express.js + OpenRouter API</p>
115     <span class="version">v1.0.0</span>
116   </div>
117 </aside>
118
119 <!-- Sidebar Overlay -->
120 <div class="sidebar-overlay" id="sidebarOverlay"></div>
121
122 <!-- Main Chat Workspace -->
123 <main class="chat-main">
124
125   <!-- Top Navigation Bar -->
126   <header class="chat-header">
127     <div class="chat-header-left">
128       <button class="btn btn-secondary btn-sm toggle-sidebar-btn" id="toggleSidebarBtn" title="Buka /
Sembunyikan Panel Pengaturan">
129         <i class="fa-solid fa-gear"></i>
130         <span>Pengaturan</span>
131       </button>
132
133       <div class="model-badge">
134         <span class="pulse-dot"></span>
135         <span id="activeModelLabel">liquid/lfm-2.5-2.6b:free</span>
136       </div>
137     </div>
138
139     <div class="chat-header-right">
140       <div class="api-key-pill" id="apiKeyPill">
141         <i class="fa-solid fa-key"></i>
142         <span id="apiKeyPillText">API Key Belum Diatur</span>
143       </div>
144       <button class="btn btn-icon" id="quickClearBtn" title="Kosongkan Chat">
145         <i class="fa-solid fa-broom"></i>
146       </button>
147     </div>
148   </header>
149
150   <!-- Warning Banner if API key missing -->
151   <div class="banner-warning" id="apiKeyBanner">
152     <i class="fa-solid fa-circle-exclamation"></i>
153     <span>API Key belum dimasukkan. Silakan masukkan OpenRouter API Key Anda di Pengaturan untuk memu
lai chat.</span>
154     <button class="btn btn-sm btn-outline-light" id="openSettingsFromBanner">
155       <i class="fa-solid fa-gear"></i> Buka Pengaturan
156     </button>
157   </div>
158
159   <!-- Messages Scroll Area -->
160   <div class="chat-body" id="chatBody">
161
162     <!-- Welcome Hero Card (Diperlihatkan jika chat belum ada) -->
163     <div class="welcome-hero" id="welcomeHero">
164       <div class="welcome-icon">
165         <i class="fa-solid fa-robot"></i>
166       </div>
167       <h1>Selamat Datang di Express AI Chatbox</h1>
168       <p>Didukung oleh model <strong>Liquid LFM 2.5 2.6B Free</strong> via OpenRouter API.</p>
169
170       <div class="suggestions-grid">
171         <button class="suggestion-card" data-prompt="Jelaskan apa itu Express.js secara singkat dan b
erikan contoh kode sederhana.">
172           <i class="fa-solid fa-code"></i>
173           <span>Contoh Express.js</span>
174         </button>
175         <button class="suggestion-card" data-prompt="Buatkan fungsi JavaScript untuk memfilter array
angka genap.">

```

```

176         <i class="fa-solid fa-laptop-code"></i>
177         <span>Fungsi JavaScript</span>
178     </button>
179     <button class="suggestion-card" data-prompt="Apa saja keunggulan model Liquid LFM 2.5?">
180         <i class="fa-solid fa-lightbulb"></i>
181         <span>Tentang Liquid LFM</span>
182     </button>
183     <button class="suggestion-card" data-prompt="Buatkan contoh puisi singkat tentang teknologi d
an kecerdasan buatan.">
184         <i class="fa-solid fa-pen-nib"></i>
185         <span>Puisi Teknologi</span>
186     </button>
187 </div>
188 </div>
189
190 <!-- Dynamic Messages container -->
191 <div class="messages-container" id="messagesContainer"></div>
192
193 <!-- Typing Indicator Loader -->
194 <div class="typing-indicator" id="typingIndicator">
195     <div class="avatar ai-avatar">
196         <i class="fa-solid fa-droplet"></i>
197     </div>
198     <div class="typing-bubble">
199         <span class="dot"></span>
200         <span class="dot"></span>
201         <span class="dot"></span>
202     </div>
203 </div>
204
205 </div>
206
207 <!-- Bottom Chat Input Controls -->
208 <footer class="chat-footer">
209     <form class="chat-form" id="chatForm">
210         <div class="input-container">
211             <textarea
212                 id="userPromptInput"
213                 placeholder="Tulis pesan Anda di sini... (Tekan Enter untuk mengirim, Shift+Enter untuk bar
is baru)"
214                 rows="1"
215                 required
216             ></textarea>
217             <button type="submit" class="send-btn" id="sendBtn" title="Kirim Pesan">
218                 <i class="fa-solid fa-paper-plane"></i>
219             </button>
220         </div>
221     </form>
222     <div class="footer-note">
223         <span>AI dapat membuat kesalahan. Periksa kembali informasi penting.</span>
224     </div>
225 </footer>
226
227 </main>
228 </div>
229
230 <!-- External Libraries: Marked.js for Markdown & Highlight.js for Syntax Highlighting -->
231 <script src="https://cdnjs.cloudflare.com/ajax/libs/marked/12.0.0/marked.min.js"></script>
232 <script src="https://cdnjs.cloudflare.com/ajax/libs/highlight.js/11.9.0/highlight.min.js"></script>
233
234 <!-- Custom Client Script -->
235 <script src="app.js"></script>
236 </body>
237 </html>
238

```

```
server.js Node.js / Express

1  const express = require('express');
2  const cors = require('cors');
3  const axios = require('axios');
4  const path = require('path');
5  require('dotenv').config();
6
7  const app = express();
8  const PORT = process.env.PORT || 3000;
9
10 // Middleware
11 app.use(cors());
12 app.use(express.json({ limit: '10mb' }));
13 app.use(express.static(path.join(__dirname, 'public')));
14
15 // OpenRouter Base Endpoint
16 const OPENROUTER_API_URL = 'https://openrouter.ai/api/v1/chat/completions';
17 const DEFAULT_MODEL = process.env.DEFAULT_MODEL || 'liquid/lfm-2.5-2.6b:free';
18
19 // Health Check Endpoint
20 app.get('/api/health', (req, res) => {
21   res.json({
22     status: 'ok',
23     timestamp: new Date().toISOString(),
24     defaultModel: DEFAULT_MODEL,
25     hasServerApiKey: Boolean(process.env.OPENROUTER_API_KEY)
26   });
27 });
28
29 // Chat API Endpoint
30 app.post('/api/chat', async (req, res) => {
31   try {
32     const { messages, apiKey: clientApiKey, model, systemPrompt } = req.body;
33
34     // Prioritaskan API Key dari Client UI (Ops 2), jika tidak ada baru gunakan process.env
35     const apiKey = clientApiKey || process.env.OPENROUTER_API_KEY;
36
37     if (!apiKey || apiKey.trim() === '') {
38       return res.status(400).json({
39         error: {
40           message: 'OpenRouter API Key belum diatur. Silakan masukkan API Key Anda di menu Pengaturan (U
41 I) atau di file .env.'
42         }
43       });
44
45       if (!messages || !Array.isArray(messages) || messages.length === 0) {
46         return res.status(400).json({
47           error: {
48             message: 'Format pesan tidak valid atau kosong.'
49           }
50         });
51       }
52
53       // Susun payload pesan dengan System Prompt jika ada
54       const formattedMessages = [];
55       if (systemPrompt && systemPrompt.trim() !== '') {
56         formattedMessages.push({
57           role: 'system',
58           content: systemPrompt.trim()
59         });
60       }
61
```

```

62 // Tambahkan percakapan user & assistant
63 messages.forEach((msg) => {
64   formattedMessages.push({
65     role: msg.role === 'user' ? 'user' : 'assistant',
66     content: msg.content
67   });
68 });
69
70 const activeModel = model || DEFAULT_MODEL;
71
72 // Kirim request ke OpenRouter API
73 const response = await axios.post(
74   OPENROUTER_API_URL,
75   {
76     model: activeModel,
77     messages: formattedMessages,
78     temperature: 0.7,
79     max_tokens: 2048
80   },
81   {
82     headers: {
83       'Authorization': `Bearer ${apiKey.trim()}`,
84       'HTTP-Referer': `http://localhost:${PORT}`,
85       'X-Title': 'Express AI Chatbox App',
86       'Content-Type': 'application/json'
87     },
88     timeout: 60000 // 60 seconds timeout
89   }
90 );
91
92 // Ambil jawaban AI dari response OpenRouter
93 const choice = response.data?.choices?.[0];
94 const aiMessage = choice?.message?.content || 'Tidak ada balasan dari model AI.';
95
96 return res.json({
97   success: true,
98   message: aiMessage,
99   model: activeModel,
100   usage: response.data?.usage || null
101 });
102
103 } catch (error) {
104   console.error('Error saat menghubungi OpenRouter API:', error.response?.data || error.message);
105
106   const statusCode = error.response?.status || 500;
107   let errorMessage = error.response?.data?.error?.message ||
108     error.response?.data?.message ||
109     error.message ||
110     'Terjadi kesalahan internal pada server.';
111
112   // Penanganan khusus jika API key salah / tidak terdaftar di OpenRouter
113   if (errorMessage.includes('User not found') || statusCode === 401) {
114     errorMessage = 'API Key OpenRouter tidak valid atau tidak ditemukan. Pastikan Anda menyalin key dari https://openrouter.ai/keys (format diawali dengan `sk-or-v1-...`).'
115   }
116
117   return res.status(statusCode).json({
118     error: {
119       message: errorMessage,
120       status: statusCode
121     }
122   });
123 }
124 });
125
126 // Fallback route untuk SPA/HTML

```



```

127 app.get('*', (req, res) => {
128   res.sendFile(path.join(__dirname, 'public', 'index.html'));
129 });
130
131 // Start Server
132 app.listen(PORT, () => {
133   console.log(`=====`);
134   console.log(`🚀 Express AI Chatbox Server berjalan di port ${PORT}`);
135   console.log(`🌐 Buka di browser: http://localhost:${PORT}`);
136   console.log(`📄 Model Default: ${DEFAULT_MODEL}`);
137   console.log(`=====`);
138 });
139

```

public/app.js JavaScript (ES6+)

```

1 /**
2  * Express AI Chatbox - Client Application Logic
3  * Integrasi OpenRouter API model: liquid/lfm-2.5-2.6b:free
4  */
5
6 document.addEventListener('DOMContentLoaded', () => {
7   // DOM Elements
8   const sidebar = document.getElementById('sidebar');
9   const sidebarOverlay = document.getElementById('sidebarOverlay');
10  const toggleSidebarBtn = document.getElementById('toggleSidebarBtn');
11  const closeSidebarBtn = document.getElementById('closeSidebarBtn');
12
13  const apiKeyInput = document.getElementById('apiKeyInput');
14  const toggleApiKeyVisibility = document.getElementById('toggleApiKeyVisibility');
15  const saveApiKeyBtn = document.getElementById('saveApiKeyBtn');
16  const apiKeyStatusDot = document.getElementById('apiKeyStatusDot');
17  const apiKeyPill = document.getElementById('apiKeyPill');
18  const apiKeyPillText = document.getElementById('apiKeyPillText');
19  const apiKeyBanner = document.getElementById('apiKeyBanner');
20  const openSettingsFromBanner = document.getElementById('openSettingsFromBanner');
21
22  const modelSelect = document.getElementById('modelSelect');
23  const activeModelLabel = document.getElementById('activeModelLabel');
24  const systemPromptInput = document.getElementById('systemPromptInput');
25  const resetSystemPromptBtn = document.getElementById('resetSystemPromptBtn');
26
27  const clearChatBtn = document.getElementById('clearChatBtn');
28  const quickClearBtn = document.getElementById('quickClearBtn');
29
30  const chatBody = document.getElementById('chatBody');
31  const welcomeHero = document.getElementById('welcomeHero');
32  const messagesContainer = document.getElementById('messagesContainer');
33  const typingIndicator = document.getElementById('typingIndicator');
34
35  const chatForm = document.getElementById('chatForm');
36  const userPromptInput = document.getElementById('userPromptInput');
37  const sendBtn = document.getElementById('sendBtn');
38  const suggestionCards = document.querySelectorAll('.suggestion-card');
39
40  // Application State
41  let apiKey = localStorage.getItem('openrouter_api_key') || '';
42  let selectedModel = localStorage.getItem('selected_model') || 'liquid/lfm-2.5-2.6b:free';
43  let systemPrompt = localStorage.getItem('system_prompt') || systemPromptInput.value;
44  let chatHistory = JSON.parse(localStorage.getItem('chat_history') || '[]');
45
46  // Configure Marked.js options
47  marked.setOptions({

```

```
48     gfm: true,
49     breaks: true,
50     highlight: function(code, lang) {
51       if (lang && hljs.getLanguage(lang)) {
52         try {
53           return hljs.highlight(code, { language: lang }).value;
54         } catch (err) {}
55       }
56       return hljs.highlightAuto(code).value;
57     }
58   });
59
60   // --- Initial Setup & UI State Sync ---
61   function init() {
62     // Populate form elements from state
63     apiKeyInput.value = apiKey;
64     modelSelect.value = selectedModel;
65     activeModelLabel.textContent = selectedModel;
66     systemPromptInput.value = systemPrompt;
67
68     updateApiKeyUI();
69     renderChatHistory();
70     setupEventListeners();
71
72     // Jika API Key belum diatur saat pertama kali dibuka, otomatis buka sidebar pengaturan
73     if (!apiKey || apiKey.trim() === '') {
74       openSidebar();
75     }
76   }
77
78   // --- Update API Key UI Indicators ---
79   function updateApiKeyUI() {
80     const hasKey = Boolean(apiKey && apiKey.trim().length > 0);
81
82     if (hasKey) {
83       apiKeyStatusDot.classList.add('connected');
84       apiKeyPill.classList.add('active');
85       apiKeyPillText.textContent = 'API Key Aktif';
86       apiKeyBanner.style.display = 'none';
87     } else {
88       apiKeyStatusDot.classList.remove('connected');
89       apiKeyPill.classList.remove('active');
90       apiKeyPillText.textContent = 'API Key Belum Diatur';
91       apiKeyBanner.style.display = 'flex';
92     }
93   }
94
95   // --- Sidebar Drawer Controls ---
96   function openSidebar() {
97     sidebar.classList.add('open');
98     sidebarOverlay.classList.add('active');
99   }
100
101   function closeSidebar() {
102     sidebar.classList.remove('open');
103     sidebarOverlay.classList.remove('active');
104   }
105
106   function toggleSidebar() {
107     if (sidebar.classList.contains('open')) {
108       closeSidebar();
109     } else {
110       openSidebar();
111     }
112   }
113
```

```

114 // --- Auto Resize Textarea ---
115 function autoResizeTextarea() {
116     userPromptInput.style.height = 'auto';
117     userPromptInput.style.height = Math.min(userPromptInput.scrollHeight, 150) + 'px';
118 }
119
120 // --- Render Chat Messages ---
121 function renderChatHistory() {
122     messagesContainer.innerHTML = '';
123
124     if (chatHistory.length === 0) {
125         welcomeHero.style.display = 'flex';
126         messagesContainer.style.display = 'none';
127     } else {
128         welcomeHero.style.display = 'none';
129         messagesContainer.style.display = 'flex';
130
131         chatHistory.forEach(msg => {
132             appendMessageUI(msg.role, msg.content, msg.timestamp, false);
133         });
134
135         scrollToBottom();
136     }
137 }
138
139 // --- Append Single Message to UI ---
140 function appendMessageUI(role, content, timestamp = new Date().toLocaleTimeString([], { hour: '2-digi
t', minute: '2-digit' })), animate = true) {
141     welcomeHero.style.display = 'none';
142     messagesContainer.style.display = 'flex';
143
144     const messageItem = document.createElement('div');
145     messageItem.className = `message-item ${role}`;
146     if (!animate) messageItem.style.animation = 'none';
147
148     const avatarHtml = role === 'user'
149       ? `<div class="avatar user-avatar"><i class="fa-solid fa-user"></i></div>`
150       : `<div class="avatar ai-avatar"><i class="fa-solid fa-droplet"></i></div>`;
151
152     // Process Markdown for AI messages
153     const formattedContent = role === 'assistant' ? processMarkdown(content) : escapeHtml(content);
154
155     messageItem.innerHTML = `
156       ${avatarHtml}
157       <div class="message-bubble-wrapper">
158         <div class="message-bubble">${formattedContent}</div>
159         <div class="message-meta">
160           <span>${role === 'user' ? 'Anda' : 'Liquid AI'}</span> • <span>${timestamp}</span>
161         </div>
162       </div>
163     `;
164
165     messagesContainer.appendChild(messageItem);
166     attachCopyCodeListeners(messageItem);
167     scrollToBottom();
168 }
169
170 // --- Process Markdown & Format Code Blocks with Copy Buttons ---
171 function processMarkdown(rawText) {
172     let html = marked.parse(rawText);
173
174     const tempDiv = document.createElement('div');
175     tempDiv.innerHTML = html;
176
177     const preElements = tempDiv.querySelectorAll('pre');
178     preElements.forEach((pre) => {

```

```
179     const codeElem = pre.querySelector('code');
180     let lang = 'code';
181
182     if (codeElem) {
183         const classNames = codeElem.className.split(' ');
184         classNames.forEach(cls => {
185             if (cls.startsWith('language-')) {
186                 lang = cls.replace('language-', '');
187             }
188         });
189     }
190
191     const codeHeader = document.createElement('div');
192     codeHeader.className = 'code-header';
193     codeHeader.innerHTML = `
194         <span><i class="fa-solid fa-code"></i> ${lang}</span>
195         <button class="copy-btn"><i class="fa-regular fa-copy"></i> Salin</button>
196     `;
197
198     pre.insertBefore(codeHeader, pre.firstChild);
199 });
200
201 return tempDiv.innerHTML;
202 }
203
204 // --- Attach Copy Listener to Code Snippets ---
205 function attachCopyCodeListeners(container) {
206     const copyBtns = container.querySelectorAll('.copy-btn');
207     copyBtns.forEach(btn => {
208         btn.addEventListener('click', () => {
209             const pre = btn.closest('pre');
210             const code = pre.querySelector('code')?.innerText || '';
211
212             navigator.clipboard.writeText(code).then(() => {
213                 btn.innerHTML = `<i class="fa-solid fa-check"></i> Tersalin!`;
214                 setTimeout(() => {
215                     btn.innerHTML = `<i class="fa-regular fa-copy"></i> Salin`;
216                 }, 2000);
217             }).catch(err => {
218                 console.error('Gagal menyalin:', err);
219             });
220         });
221     });
222 }
223
224 // --- Escape Raw HTML for User Safety ---
225 function escapeHtml(text) {
226     const div = document.createElement('div');
227     div.textContent = text;
228     return div.innerHTML.replace(/\n/g, '<br>');
229 }
230
231 // --- Scroll Chat to Bottom ---
232 function scrollToBottom() {
233     chatBody.scrollTo({
234         top: chatBody.scrollHeight,
235         behavior: 'smooth'
236     });
237 }
238
239 // --- Send Message Handler ---
240 async function handleSendMessage() {
241     const promptText = userInput.value.trim();
242     if (!promptText) return;
243
244     // Check API Key
```

```

245     if (!apiKey || apiKey.trim() === '') {
246         openSidebar();
247         apiKeyInput.focus();
248         alert('Silakan masukkan OpenRouter API Key Anda terlebih dahulu di menu Pengaturan!');
249         return;
250     }
251
252     // Collapse sidebar saat mulai mengirim pesan agar layar lebih luas
253     closeSidebar();
254
255     // Append User Message to State & UI
256     const timeNow = new Date().toLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
257     const userMessage = { role: 'user', content: promptText, timestamp: timeNow };
258     chatHistory.push(userMessage);
259     localStorage.setItem('chat_history', JSON.stringify(chatHistory));
260
261     appendMessageUI('user', promptText, timeNow);
262
263     // Reset Input
264     userPromptInput.value = '';
265     autoResizeTextarea();
266
267     // Show Typing Indicator
268     typingIndicator.classList.add('active');
269     sendBtn.disabled = true;
270     scrollToBottom();
271
272     try {
273         const apiMessagesPayload = chatHistory.map(item => ({
274             role: item.role,
275             content: item.content
276         }));
277
278         const response = await fetch('/api/chat', {
279             method: 'POST',
280             headers: {
281                 'Content-Type': 'application/json'
282             },
283             body: JSON.stringify({
284                 messages: apiMessagesPayload,
285                 apiKey: apiKey.trim(),
286                 model: selectedModel,
287                 systemPrompt: systemPrompt
288             })
289         });
290
291         const data = await response.json();
292
293         if (!response.ok) {
294             throw new Error(data.error?.message || 'Terjadi kesalahan saat menghubungkan ke AI.');
```

```

311     appendMessageUI('assistant', errorMessage, errTime);
312   } finally {
313     typingIndicator.classList.remove('active');
314     sendBtn.disabled = false;
315     scrollToBottom();
316   }
317 }
318 }
319
320 // --- Event Listeners Setup ---
321 function setupEventListeners() {
322   // Sidebar Toggles
323   if (toggleSidebarBtn) toggleSidebarBtn.addEventListener('click', toggleSidebar);
324   if (closeSidebarBtn) closeSidebarBtn.addEventListener('click', closeSidebar);
325   if (sidebarOverlay) sidebarOverlay.addEventListener('click', closeSidebar);
326
327   if (openSettingsFromBanner) {
328     openSettingsFromBanner.addEventListener('click', () => {
329       openSidebar();
330       apiKeyInput.focus();
331     });
332   }
333
334   // Save API Key
335   saveApiKeyBtn.addEventListener('click', () => {
336     apiKey = apiKeyInput.value.trim();
337     localStorage.setItem('openrouter_api_key', apiKey);
338     updateApiKeyUI();
339
340     saveApiKeyBtn.innerHTML = `<i class="fa-solid fa-check"></i> Tersimpan!`;
341     setTimeout(() => {
342       saveApiKeyBtn.innerHTML = `<i class="fa-solid fa-floppy-disk"></i> Simpan Key`;
343       closeSidebar();
344     }, 1200);
345   });
346
347   // Toggle Eye Password
348   toggleApiKeyVisibility.addEventListener('click', () => {
349     const type = apiKeyInput.getAttribute('type') === 'password' ? 'text' : 'password';
350     apiKeyInput.setAttribute('type', type);
351     toggleApiKeyVisibility.innerHTML = type === 'password'
352       ? `<i class="fa-solid fa-eye"></i>`
353       : `<i class="fa-solid fa-eye-slash"></i>`;
354   });
355
356   // Change Model
357   modelSelect.addEventListener('change', (e) => {
358     selectedModel = e.target.value;
359     localStorage.setItem('selected_model', selectedModel);
360     activeModelLabel.textContent = selectedModel;
361   });
362
363   // System Prompt Editor
364   systemPromptInput.addEventListener('input', (e) => {
365     systemPrompt = e.target.value;
366     localStorage.setItem('system_prompt', systemPrompt);
367   });
368
369   resetSystemPromptBtn.addEventListener('click', () => {
370     const defaultPrompt = 'Kamu adalah Liquid AI, asisten virtual yang cerdas, ramah, dan sangat membantu dalam bahasa Indonesia.';
371     systemPrompt = defaultPrompt;
372     systemPromptInput.value = defaultPrompt;
373     localStorage.setItem('system_prompt', defaultPrompt);
374   });
375

```

```
376 // Clear Chat History
377 const clearChatHandler = () => {
378   if (confirm('Apakah Anda yakin ingin menghapus seluruh riwayat chat?')) {
379     chatHistory = [];
380     localStorage.removeItem('chat_history');
381     renderChatHistory();
382     closeSidebar();
383   }
384 };
385
386 clearChatBtn.addEventListener('click', clearChatHandler);
387 quickClearBtn.addEventListener('click', clearChatHandler);
388
389 // Auto resize input & keyboard shortcuts
390 userPromptInput.addEventListener('input', autoResizeTextarea);
391 userPromptInput.addEventListener('keydown', (e) => {
392   if (e.key === 'Enter' && !e.shiftKey) {
393     e.preventDefault();
394     handleSendMessage();
395   }
396 });
397
398 // Form submit
399 chatForm.addEventListener('submit', (e) => {
400   e.preventDefault();
401   handleSendMessage();
402 });
403
404 // Quick Suggestions
405 suggestionCards.forEach(card => {
406   card.addEventListener('click', () => {
407     const prompt = card.getAttribute('data-prompt');
408     if (prompt) {
409       userPromptInput.value = prompt;
410       autoResizeTextarea();
411       userPromptInput.focus();
412     }
413   });
414 });
415 }
416
417 // Start App
418 init();
419 });
420
```